

Structure-Based Robust Fractal Graph Neural Network with Molecular Fingerprint BERT for Molecular Property Prediction

Yaguo Dong¹, Meiling Xu¹, *Member, IEEE*, and Lixin Tang², *Fellow, IEEE*

Abstract—Accurate molecular property prediction is of great importance for AI-based drug design and bioinformatics. Despite recent promising progress, existing methods face challenges due to insufficient data labeling and imbalanced label distribution. Moreover, many current molecular property prediction models are based on 2D molecular graphs, neglecting the 3D characteristics of molecules. Therefore, in this paper, we propose the structure-based robust fractal graph neural network with Molecular Fingerprint Bidirectional Encoder Representations from Transformers (MF-BERT) model for molecular property prediction. We construct the pre-trained MFBERT model to explore the use of unlabeled data and obtain representations from molecular extended connectivity fingerprints as the node features of molecular graphs. Then we design the fractal graph neural network that incorporates molecular 3D conformations while ensuring equivariance to molecular rotations and translations. A 3D feature extraction module and a coordinate update module are proposed to enhance the model's ability to capture molecular structural features. Additionally, weighted-gradient distributionally robust optimization is combined with the gradient descent algorithm to update model parameters, which can mitigate issues stemming from imbalanced label distribution and improve the robustness of our model. We conducted experiments on seven datasets from MoleculeNet, comprising three regression tasks and four classification tasks. Our method achieves the state-of-the-art performance compared to the existing models on all these datasets, which demonstrates its effectiveness for molecular property prediction. Additionally, we conducted ablation experiments on our designed modules, further validating the necessity of each component in our method.

Index Terms—Molecular property prediction, fractal graph neural network, molecular fingerprint BERT, distributionally robust optimization, molecular 3D conformation.

I. INTRODUCTION

MOLECULAR Property Prediction (MPP) [1], [2] aims to predict their physicochemical properties or biological activities under specific conditions by analyzing the molecular structures and characteristics with computational methods or machine learning techniques. Molecular property prediction holds significant importance in the fields of chemistry, materials science, drug discovery and drug design. By predicting the molecular properties, researchers can understand their behaviors, expedite the discovery process of new materials and drugs, reduce experimental time and costs, and enhance the success rate.

Traditional methods for predicting molecular properties primarily depend on quantum chemistry calculations [3] and molecular dynamics simulations [4], [5]. While these methods achieve high accuracy, they typically require substantial computational resources and time, especially for large molecular systems. Due to the high computational cost, they are not suitable for large-scale molecular library screening. Therefore, developing efficient and accurate methods for molecular property prediction has become a major focus and challenge in current research.

Recently, with the rapid advancement of Artificial Intelligence (AI) technologies [6], [7], [8], machine learning and deep learning methods have brought new prospects for molecular property prediction. They can learn relationships between molecular features or structures and properties from large-scale datasets, enabling the rapid and accurate prediction for unknown molecules. Particularly with the development of deep learning, neural network-based models have demonstrated outstanding performance in MPP. By automatically extracting molecular features, these models can handle complex molecular structures and exhibit strong generalization capabilities.

Recurrent Neural Networks (RNNs) [9] and Convolutional Neural Networks (CNNs) [10] have been employed to process one-dimensional sequence data of molecules for property prediction tasks. Various Graph Neural Network (GNN) models [11], [12], such as Graph Convolutional Networks (GCNs) [13], Graph Attention Networks (GATs) [14] and Graph Isomorphism Network (GIN) [15], have been designed to handle

Received 19 August 2024; revised 24 November 2024; accepted 17 January 2025. Date of publication 15 April 2025; date of current version 16 January 2026. This work was supported by the Major Program of National Natural Science Foundation of China under Grant 72192830, and Grant 72192831, in part by the National Natural Science Foundation of China under Grant 62176049, and in part by the 111 Project under Grant B16009. (Corresponding author: Lixin Tang.)

Yaguo Dong is with the National Frontiers Science Center for Industrial Intelligence and Systems Optimization, Northeastern University, Shenyang 110819, China, and also with the Key Laboratory of Data Analytics and Optimization for Smart Industry (Northeastern University), Ministry of Education, Shenyang 110819, China (e-mail: 2110321@stu.neu.edu.cn).

Meiling Xu is with the Liaoning Engineering Laboratory of Data Analytics and Optimization for Smart Industry, Shenyang 110819, China, and also with the Liaoning Key Laboratory of Manufacturing System and Logistics Optimization, Shenyang 110819, China (e-mail: xumeiling@mail.neu.edu.cn).

Lixin Tang is with the National Frontiers Science Center for Industrial Intelligence and Systems Optimization, Northeastern University, Shenyang 110819, China (e-mail: lixintang@mail.neu.edu.cn).

Recommended for acceptance by Z. Zhu.

Digital Object Identifier 10.1109/TETCI.2025.3543308

the two-dimensional molecular graph structures, thereby improving property prediction. These GNN models [16] excel at capturing the intricate relationships between atoms and bonds in a molecule, allowing for more accurate and robust predictions of molecular properties.

However, in the field of molecular property prediction, several key challenges persist. Firstly, despite advancements in experimental techniques leading to an accumulation of substantial molecular data, there is little labeled data for molecular properties and the data is usually unlabeled. Therefore, how to effectively extract pertinent features from these unlabeled data to aid in molecular property prediction remains a significant challenge. Secondly, existing labeled datasets typically exhibit imbalanced distribution on various molecular properties, thereby compromising the predictive capabilities of models towards minority classes during training. Finally, while the graph-based models are extensively employed in molecular property prediction, the effective utilization of 3D molecular structural characteristics is often neglected.

To address these challenges, we propose the structure-based robust fractal graph neural network combined with the pre-trained Molecular Fingerprint Bidirectional Encoder Representations from Transformers (MFBERT) model for molecular property prediction. Based on the similarity between sentences of natural language and molecular sequence representation, the substructures derived from molecular Extended Connectivity Fingerprints (ECFP) [17] can be analogized to words. Then our MFBERT model is proposed in this paper to obtain molecular ECFP representations from the unlabeled data and alleviate the problem of insufficient labeled data in molecular property prediction tasks. Moreover, we design a Fractal Graph Neural Network (FGNN) to incorporate the 3D characteristics with a 3D feature extraction module and a coordinate update module. Our FGNN model can extract features of molecular different scales with the fractal framework and ensure the equivariance of molecular rotations and translations with 3D conformations. Finally, the Weighted-gradient Distributionally Robust Optimization (W-DRO) is combined with gradient descent algorithm in the training process to solve the problem of imbalanced distribution, which improves the model's generalization and robustness.

The key contributions of this paper are summarized as follows:

- The pre-trained model MFBERT is proposed to explore the utilization of unlabeled molecular data and extract the molecular ECFP embeddings as the node features of molecular graphs.
- We incorporated molecular 3D characteristics by the designed fractal graph neural network to enhance the model's ability of extracting structural features and ensure the equivariance of molecules with respect to rotation, translation, and permutation, which improves the prediction performance.
- We combined the improved distributionally robust optimization method with the gradient descent algorithm to mitigate the issue of data imbalance in prediction tasks.
- The effectiveness of our proposed method is validated through experiments on seven datasets. Our results

achieved the improvements on all molecular property prediction tasks compared to the state-of-the-art methods.

The remaining sections of this paper are structured as follows. The related works of molecular property prediction are summarized in Section II, including traditional MPP methods, machine learning-based MPP methods, and deep learning-based MPP methods. Section III presents our proposed method, including molecular fingerprint BERT model, our designed fractal graph neural network and the optimization algorithm combined with W-DRO. Section IV describes seven benchmark datasets, the evaluation metric, implementation details, and our experimental results. Finally, we draw the conclusions of this paper in Section V.

II. RELATED WORK

Effective molecular representation and accurate property prediction are critical tasks in the workflow of AI-based drug design. Accurate prediction methods can significantly accelerate the identification of better drug candidates, making the process faster and more cost-effective. Existing frameworks for molecular property prediction can be classified into three categories: traditional MPP methods, machine learning-based MPP methods and deep learning-based MPP methods.

A. Traditional Molecular Property Prediction Methods

Traditional MPP methods include empirical formula methods, quantum mechanical calculations and molecular dynamics simulations. Molecular property prediction evolved from Structure-Activity Relationship (SAR) analysis, based on the “lock-and-key” hypothesis [18] proposed by Paul Ehrlich. This hypothesis suggests that the interaction between a compound (key) and a biological target (lock) depends on their spatial compatibility. However, SAR methods largely relied on the chemist's experience and intuition, lacking a rigorous theoretical foundation and standardized analytical methods. To overcome these limitations, researchers began using mathematical and statistical methods to quantitatively analyze the relationships between molecular structures and biological activities. In 1868, Alexander and Thomas proposed the Crum-Brown equation [19], which laid the groundwork for Quantitative Structure-Activity Relationship (QSAR) studies. In 1994, Hansch proposed the QSAR model [20], which predicts molecular properties based on the relationship between chemical structures and experimental data.

For quantum mechanical calculations, the Hartree-Fock method and Density Functional Theory (DFT) are commonly used. In 1951, John Slater and John C. Gordon applied the Hartree-Fock method [3] to chemistry, calculating the ground state energy and electronic structure of various molecules. On the other hand, DFT [21] assumes that quantum interactions between particles determine the internal correlation within a molecule, which in turn determines its intrinsic properties. According to this theory, DFT replaces the wave function with electron density as the basis for research and solves the energy of entire molecular system, enabling the direct calculation of various molecular and material properties from the ground state electron density. In molecular dynamics simulations, researchers

use tools like GROMACS [4] and AMBER [5] for a variety of molecular simulation applications, providing in-depth studies of the structure, dynamics and interactions of biological macromolecules.

B. Machine Learning-Based Molecular Property Prediction Methods

With the advancements of technology, researchers have increasingly combined machine learning methods with traditional fingerprints and manually designed features to achieve more accurate molecular property predictions. Machine learning-based MPP methods initially rely on human expertise to select key characteristics and generate molecular descriptors. These representations are then input into specially designed machine learning models, which are trained by minimizing a loss function until satisfactory accuracy is achieved. Common machine learning algorithms used in molecular property prediction include Random Forest (RF) [22], support vector machine [23], XGBoost and so on.

In 2003, Vladimir et al. [22] introduced RF model for MMP based on quantitative descriptions of molecular compound structures. To enhance the predictive performance of RF models, XGBoost [24] was introduced into the molecular property prediction, offering the advantage of fast computation. In 2021, Dominik et al. [25] utilized graph-to-structure model in molecular machine learning. The model leverages implicit correlations in training data to infer inter-atomic distances, bypassing traditional energy minimization and achieving high accuracy with reduced computational cost.

C. Deep Learning-Based Molecular Property Prediction Methods

Since researchers struggle with designing effective descriptors due to unclear biochemical mechanisms, leading to challenges in building accurate machine learning models, deep learning algorithms [26] are used to automatically learn task-specific features and reduce the need for complex feature engineering. These methods can be categorized based on the molecular data representation: one-dimensional (1D), two-dimensional (2D), three-dimensional (3D), and multi-modal methods.

The one-dimensional molecular data is typically represented by strings using specific syntax, such as Simplified Molecular Input Line Entry System (SMILES) [27]. Convolutional neural networks, recurrent neural networks and Transformer are usually used to model 1D sequence representations. Maya et al. [10] proposed the method of convolutional molecular fingerprints, which utilized CNNs to learn molecular features as low-dimensional vectors based on SMILES, thereby enhancing the expressive capability of sequence embedding. Esben et al. [9] improved the accuracy of MPP by enumerating SMILES to augment the SMILES molecular dataset used to train bidirectional long short-term memory networks. ChemBERTa-2 [28] is a pre-trained model based on the self-attention mechanism and built on the language of SMILES, demonstrating competitive performance on the MoleculeNet benchmark tasks by leveraging large-scale pre-training datasets and hyper-parameter tuning.

The richer structural information in 2D molecular data enhances the performance of MPP. The most common type of 2D molecular representation is the undirected graphs, where molecules are depicted using adjacency matrices and feature matrices for atoms and chemical bonds. Currently, GNNs are the primary models to extract features from molecular graphs, including the GCN [13], GIN [15] and others. These models utilize a message-passing mechanism to continuously aggregate and update messages from neighboring nodes, making them well-suited for modeling molecular graph structures. Lu et al. [29] introduced a multi-level graph convolutional neural network for molecular property prediction, aiming to provide a generalizable and transferable MPP model. Zhang et al. [30] introduced motif-based graph self-supervised learning to leverage GNNs for molecular generation and prediction tasks by focusing on frequently occurring sub-graphs in molecular graphs, which achieved promising performance.

Although 2D molecular graph representations can capture some structural information, the 3D molecular characteristics are equally important, offering more detailed insights. Introducing 3D characteristics can enrich the structural features and enhance the accuracy of predictions. Johannes et al. [31] proposed the DimeNet model, which embedded atomic spatial position information into graph neural networks, and used spherical Bessel functions and spherical harmonics to construct well-orthogonal representations, leading to more precise predictions. Fang et al. [32] proposed a Geometry-Enhanced molecular representation learning Method (GEM) to enhance the modeling of molecules using graph neural networks. Stark et al. [33] introduced 3D Infomax to enhance GNNs for molecular property prediction by pre-training models to reason about the geometry of molecules.

To obtain more comprehensive molecular representations, some researchers have attempted to integrate different-dimensional molecular representations, such as PremuNet [34] and SGGRL [35]. PremuNet [34] is a pre-trained multi-representation fusion model that integrates molecular SMILES strings, graph representations and conformations to extract comprehensive molecular information and the model obtained promising results on different molecular property prediction tasks. Zhang et al. [35] designed the multi-modal representation learning model SGGRL and used a fusion layer to integrate the molecular representations of different modalities including the sequence, graph, and geometry characteristics.

III. METHODOLOGY

Although much progress has been made in molecular property prediction, several challenges remain, such as the scarcity of labeled data, the imbalance problem in label distribution and the lack of effective utilization for 3D characteristics. To address these challenges, we propose the structure-based robust fractal graph neural network with MFBERT for molecular property prediction. As illustrated in Fig. 1, our method follows a sequential process: dataset partition, data processing, pre-training MFBERT and FGNN, molecular feature embedding with MFBERT, feature extraction with FGNN, pooling to obtain

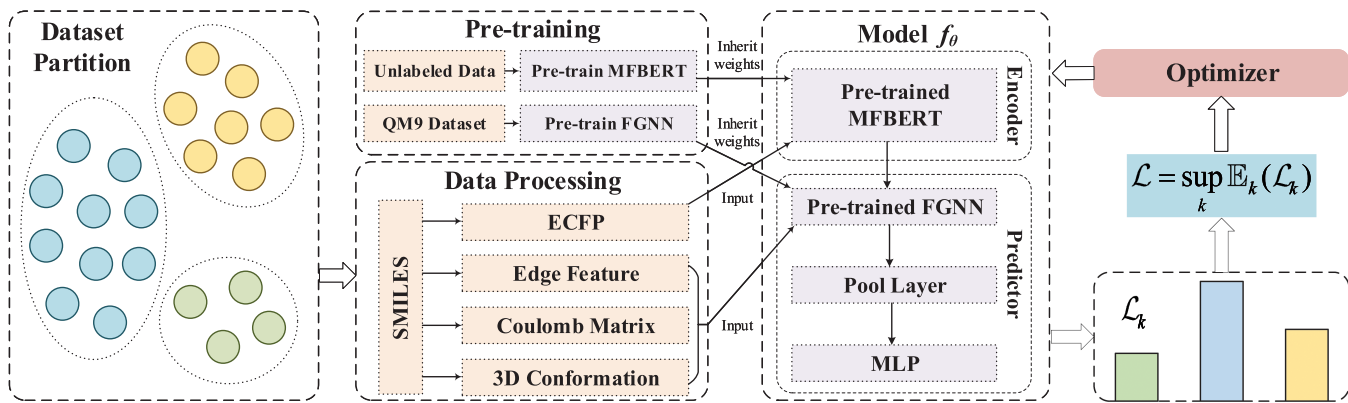


Fig. 1. The framework of our structure-based robust fractal graph neural network with molecular fingerprint BERT for the molecular property prediction task, including dataset partition, data processing, pre-training MFBERT, pre-training FGNN, molecular property prediction with MFBERT and FGNN, and the optimization process.

molecular graph features, prediction with the Multi-Layer Perceptron (MLP), and finally, updating model parameters using the gradient descent algorithm combined with weighted-gradient distributionally robust optimization.

For the data partition, we split the molecular dataset depending on their labels. In the data processing, we will calculate molecular ECFPs as the input to pre-train MFBERT model. The molecular 3D conformations, edge features and Coulomb matrix are calculated as the auxiliary information to predict its property. In this section, we will introduce our molecular property prediction method from three parts, which are molecular fingerprint BERT, fractal graph neural network and the optimization algorithm.

A. Molecular Fingerprint BERT

In bioinformatics, molecules can be represented as strings with SMILES notation. In natural language, a sentence is typically a chain of words. Based on the structural similarity between the natural language and molecular sequence representation, the relationship between sentences and molecules is established. Therefore, molecules can be processed with natural language processing methods. As shown in Fig. 2, we propose the MFBERT model to obtain the molecular features. In order to capture the atom feature and its environment information in the process of obtaining molecular embedding, ECFPs are used to generate molecular substructure sequences, which serve as the input to MFBERT.

Extended connectivity fingerprint [17] is one of the types of circular fingerprints and widely used to construct QSAR models. The core of ECFP comes from Morgan algorithm. The definition of ECFP is the set of atomic identifiers for each perception radius from 0 to the limit r . As the radius increases, the set includes all identifiers found in both previous iterations and the current one. When the radius $r = 0$, only the atoms themselves are taken into account. For the ECFP with $r = 1$, the members will contain their neighbors besides the atoms themselves. By this, the environment of atoms can be taken into account in the process of generating ECFPs. The calculation of identifiers occurs in three stages, which are the initial assignment, iterative update

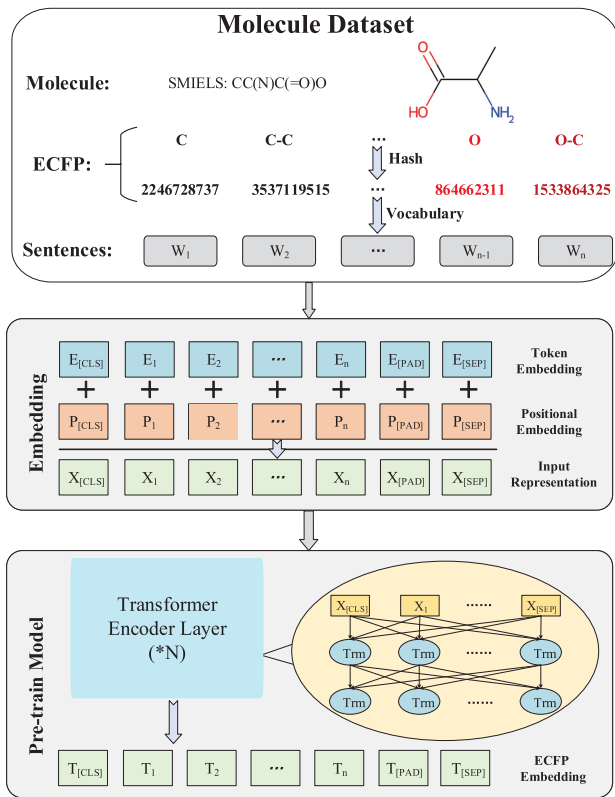


Fig. 2. The framework of our Molecular Fingerprint Bidirectional Encoder Representations from Transformers (MFBERT) model, from molecular ECFPs generation as the input to feature extracting with Transformer encoder layers.

and duplicate processing. Firstly, each atom is given an initial integer identifier, which can be the atomic number, atomic mass, etc. Secondly, in the iterative stage, the information of neighbor atoms is grouped for the atom to form an array, and then the array is hashed to a number. Finally, the identifier with the same features will be deleted. The generation process of ECFPs is shown in Algorithm 1.

When obtaining molecular ECFP representations, the sequences are input into MFBERT for pre-training to acquire the embeddings. As shown in Fig. 2, MFBERT can be divided into

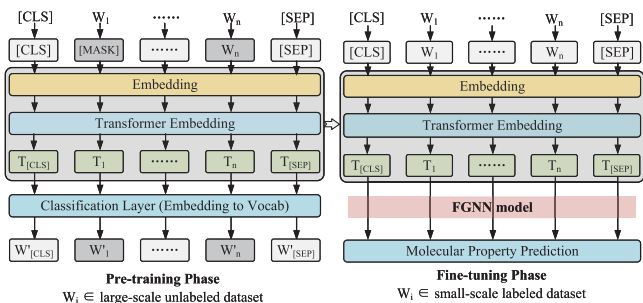


Fig. 3. Two phases for molecular property prediction with MFBERT: pre-training phase and fine-tuning phase.

Algorithm 1: ECFP Generation Algorithm

Input: A molecule M ; set S ; radius r ;

Output: Generated ECFP of M ;

- 1: Label each atom with a 32-bit integer by Morgan algorithm and hash it to obtain its identifier;
 - 2: Add the identifier of each atom to set S ;
 - 3: **if** $r > 0$ **then**
 - 4: **for** $i = 1, \dots, r$ **do**
 - 5: Create a feature list F for each atom;
 - 6: Fill F in the format $[i, \text{identifier}, \text{bo1}, \text{aid1}, \text{bo2}, \text{aid2}, \dots]$, where i is the iteration index, identifier corresponds to the current atom, boX is the order of bond X, and aidX is the identifier of the atom connected to the bond X;
 - 7: Compute the hash value of F as the new identifier for the atom;
 - 8: Add the new identifier to the set S if it is not in S ;
 - 9: **end for**
 - 10: **end if**
 - 11: Return the set S .
-

two parts: the embedding layer and the Transformer encoder layers [36]. In the embedding layer, since there is no context relationship between molecular structures, we omit the segment embedding and the input representation is the sum of the token embedding and the position embedding. Token embedding is the vector representation of molecular substructure. Position embedding represents the position of molecular substructure in ECFP sequences. The core of MFBERT is based on Transformer language model [37], and it consists of a varying number of multi-head self-attention encoder layers.

As shown in Fig. 3, there are two phases for predicting molecular properties with MFBERT. Firstly, we need to pre-train MFBERT on a large-scale unlabeled dataset to learn molecular characteristics. In this phase, only Mask Language Model (MLM) task is used as the training objective. Then the model is fine-tuned according to the task-specific labeled dataset to improve prediction performance and with the weights inherit from the pre-trained model. In the fine-tuning phase, FGNN is designed to predict the molecular property.

With MFBERT model, we can address two problems. One is that molecular substructures may play different roles in different

environments. The ECFP embedding, generated by MFBERT, can be changed dynamically based on context. The other problem is that less labeled molecular data is insufficient to train supervised deep learning models from scratch. For MPP task, our model can inherit weights from the pre-trained MFBERT model, which is trained on the large unlabeled dataset with self-supervised mode and fine-tune the model to realize the better prediction performance.

B. Fractal Graph Neural Network

Fractal [38], originating from geometry, refers to solving a problem from different levels, perspectives and components. This methodology leverages the inherent self-similarity and recursive nature of fractals to tackle complex problems in a structured and hierarchical manner. Fig. 4 is the diagram of our fractal graph neural network, which is based on the fractal expansion rule [39], [40] and can extract molecular features at different scales. As shown in Fig. 4, our FGNN block consists of sub-paths of varying lengths and enhances performance by selecting the optimal set of sub-paths. Here, the shallow network provides faster responses and the deep network yields more accurate results. The outputs of different sub-paths are combined using a joining operation. The other key in FGNN is drop-path, which is a regularization method and an extension of dropout that prevents over-fitting and improves the model performance. When multiple layers of varying depth are joined, the model randomly turns off some path inputs, while ensuring at least one input remaining.

In Fig. 4, the orange block represents the designed graph neural network and green join layer denotes the mean operator in this paper. Our molecular property prediction model based on FGNN is composed of multiple graph neural network blocks, a pooling layer and the MLP prediction layer. The inputs to FGNN include ECFP embeddings from the pre-trained MFBERT, molecular edge features, molecular 3D coordinates and the Coulomb matrix. The graph neural network layer in FGNN model is divided into four components: the feature embedding layers, 2D feature extraction module, 3D feature extraction module and the coordinate update module.

In the embedding layers, the inputs are transformed into FGNN model. The initial node features for the constructed graph neural network are extracted by averaging the ECFP embeddings of the atom and its environmental contexts across different radii. Then the embedding calculations are as follows:

$$h_i = \phi_n(h_i^0) \quad (1)$$

$$e_{ij} = \phi_e(e_{ij}^0) \quad (2)$$

$$c_i = \phi_c(\text{CM}_i), \quad (3)$$

where i and j are the indices of atoms in the molecules, corresponding to the start and end nodes of an edge. h^0 is the initial node features encoded by MFBERT model and e^0 is the predefined edge features. CM represents the molecular Coulomb matrix. ϕ_n , ϕ_e and ϕ_c represent the embedding layers. h , e and c are the embedding representation.

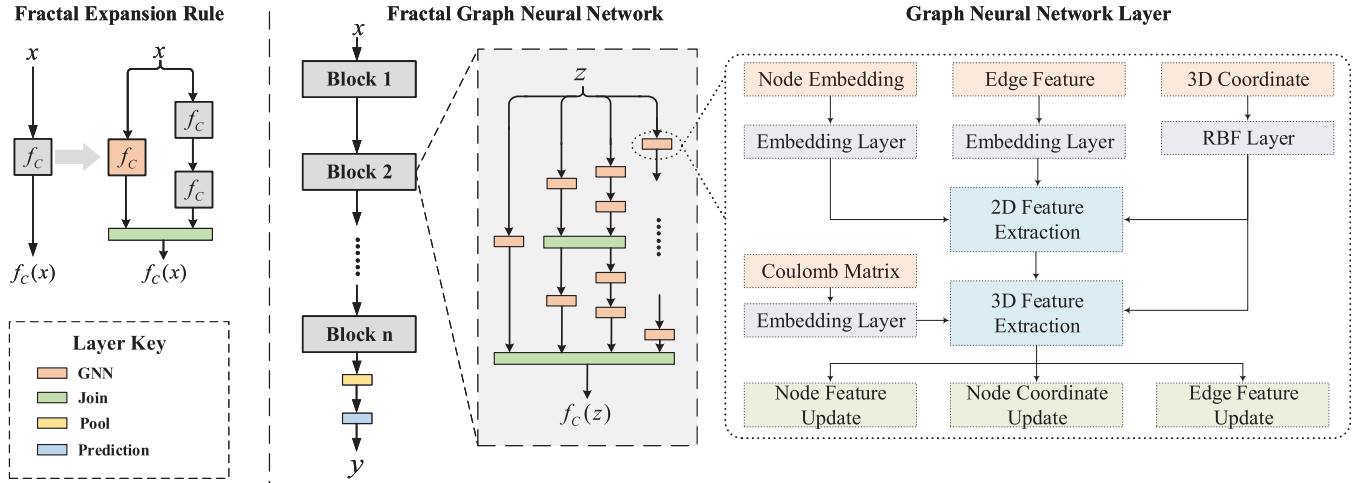


Fig. 4. The diagram of Fractal Graph Neural Network (FGNN) architecture. The diagram respectively illustrates the fractal expansion rule, the framework of fractal graph neural network and the proposed graph neural network layer.

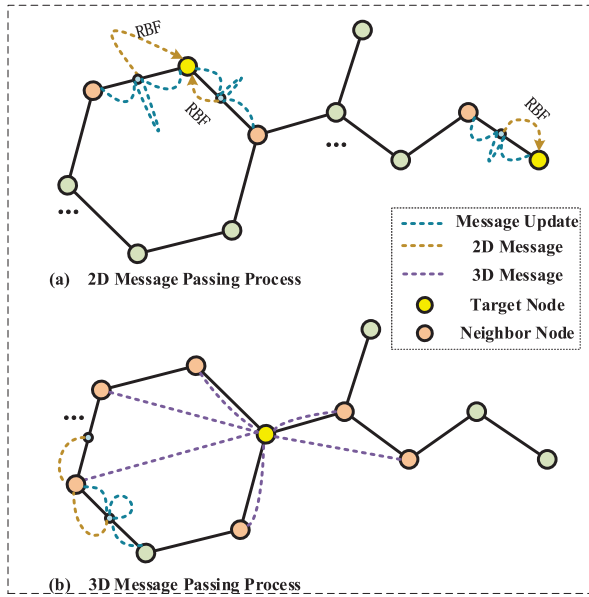


Fig. 5. The diagram of 2D message passing process and 3D message passing process.

For the input of atom coordinates, we construct the weights of messages from them with Radial Basis Function (RBF). The relative distance between atoms $d_{ij} = \|p_i - p_j\|$ is computed to maintain the molecular property of rotational invariance. Then RBF is used to expand the distances, which is as follows:

$$rbf_{i,j} = \phi_{\text{rbf}}(\exp(-\gamma \|d_{ij} - \mu\|^2)), \quad (4)$$

where p_i is the coordinate of i^{th} node, γ and μ are the parameters, ϕ_{rbf} is the transformation layer, $rbf_{i,j}$ is the weight of message from node j to node i .

In the 2D feature extraction module, we obtain the molecular features based on the molecular graph $\mathcal{G} = \{\mathcal{V}, \mathcal{E}\}$, where $\mathcal{V}$ is the node set in molecules and $\mathcal{E}$ is the chemical bond set. The node features are the transformation of MFBERT embedding. As shown in Fig. 5(a), the computation process begins with

updating the messages based on the features of starting node, ending node and the edge itself, which is depicted with a blue dashed line. This is followed by message aggregation from its neighboring nodes, and finally the node features are updated, which is presented with yellow dashed line. The calculation process is as follows:

$$m_{i,j}^{2D} = \phi_{2D}(h_i, h_j, e_{ij}) \quad (5)$$

$$m_i^{2D} = \sum_{j \in \mathcal{N}_i} m_{i,j}^{2D} * rbf_{i,j} \quad (6)$$

$$h_i = h_i + m_i^{2D}, \quad (7)$$

where $m_{i,j}^{2D}$ is the calculated 2D message between node i and j . $\mathcal{N}_i$ is the neighbor nodes of the i^{th} atom. h is the node features and e is the edge features. ϕ_{2D} is the multi-layer neural network. Then $m_{i,j}^{2D}$ will be updated to new edge features $e_{i,j}$.

To enhance the features extraction of molecules, we have incorporated the Coulomb matrix representation of the molecules into the 3D feature extraction module, leveraging the molecular coordinates to provide a more comprehensive and detailed depiction of the molecular structure. The message passing processing is shown in Fig. 5(b). To prevent the node features in the model from becoming too similar and to accelerate computation, we update the message on a randomly selected subset $\mathcal{S}_R$ of nodes rather than all nodes in the space. By stacking multiple layers, we can effectively extract the 3D features of molecules. The calculations are as follows:

$$m_{i,j}^{3D} = \phi_{3D}(h_i, h_j, c_i) \quad (8)$$

$$m_i^{3D} = \sum_{j \in \mathcal{S}_R} m_{i,j}^{3D} * rbf_{i,j} \quad (9)$$

$$h_i = h_i + m_i^{3D} \quad (10)$$

where m^{3D} is the calculated 3D message and ϕ_{3D} is the multi-layer neural network for updating 3D messages.

Moreover, we add a coordinate update module in FGNN model to further enhance structural information. By dynamically

updating the coordinates of nodes, the geometry of molecules can be more accurately captured and represented, improving the model's comprehension for spatial relationships. We use the obtained node features and original coordinates to update molecular conformations, which is:

$$p = p + \Delta_p = p + \phi_p(h, p), \quad (11)$$

where Δ_p is the change in coordinates. ϕ_p is the multi-layer neural network to predict the changes of coordinates. Then in the next layer, the new weights of messages passing and Coulomb matrix features will be updated.

Finally, we use a pooling layer and MLP network to predict the molecular property. Here, the mean pooling layer is adopted to obtain the graph feature g from the node features,

$$g = \text{readout}(h) = \frac{1}{n} \sum_i^n h_i \quad (12)$$

$$\hat{y} = \phi_o(g), \quad (13)$$

where n is the number of atoms in the molecule. ϕ_o represents MLP model and $\hat{y}$ is the predicted molecular property.

C. Optimization Algorithm

In this paper, we combined the gradient descent algorithm with weighted-gradient distributionally robust optimization [41] to update the model parameters of MFBERT and FGNN. In molecular property prediction tasks, the problem of imbalanced property label distribution is a common issue. Therefore, we formulate our property prediction problem as a Distributionally Robust Optimization (DRO) problem by dataset partition according to labels of molecular properties, which is:

$$\min_{\theta} F_{\mathbf{p}}[\theta] = \min_{\theta} \sup_{\mathbf{p}} \{ \mathbf{p}^T \mathcal{L}(f_{\theta}(x), y) - l(\mathbf{p}, \mathcal{U}) \}, \quad (14)$$

where θ is the parameters of our molecular property prediction model f . $\mathbf{p} \in \mathbb{R}^n$ represents the probability distribution of molecular data, and $\mathcal{U}$ is the uniform probabilities. $l(\mathbf{p}, \mathcal{U})$ is the measure between two distributions. $\mathcal{L}$ is the loss function between the prediction value $f_{\theta}(x)$ and target value y on the molecular dataset.

In our task, the data is obtained through uniform sampling, which makes $\mathbf{p}$ follows the uniform distribution and the term $l(\mathbf{p}, \mathcal{U})$ is 0. With the constructed partitioned dataset $\mathcal{M} = \{\mathcal{M}_1, \mathcal{M}_2, \dots, \mathcal{M}_K\}$ for supervised learning, our problem defined in 14 can be further formalized as follows:

$$F_{\mathcal{M}_k}[\theta] = \sup_{x \sim \mathcal{M}_k} \mathbb{E}(\mathcal{L}_{\mathcal{M}_k}(x)), \quad (15)$$

where $k \in \{1, 2, \dots, K\}$ and K denotes the number of subsets after dataset partition.

For the partitioned dataset $\mathcal{M}$, the average loss on each subset $\mathcal{M}_k$ is calculated and each expectation $\mathbb{E}_k$ of loss function $\mathcal{L}_k$ is linear. Then we can use the supremum of linear expectations $\mathbb{E}_k$ to represent the sublinear expectation $\hat{\mathbb{E}}$ of the loss function $\mathcal{L}$. In this context, minimizing the maximum average loss across all subsets becomes the objective of the problem optimizing the

parameters θ :

$$\begin{aligned} \min_{\theta} \hat{\mathbb{E}}_{(x,y) \sim \mathcal{M}} [\mathcal{L}(f_{\theta}(x), y)] \\ = \min_{\theta} \max_{1 \leq k \leq K} \frac{1}{m_k} \sum_{i=1}^{m_k} \left[\mathcal{L}(f_{\theta}(x_{ki}), y_{ki}) \right], \end{aligned} \quad (16)$$

where k is the index of subset and m_k denotes the sample size of k^{th} subset. Then the direction of gradient descent algorithm for optimizing the model parameters θ can be obtained by solving the problem defined in 16.

This method allows us to focus not only on the overall loss across the dataset but also on the loss for each subset. By minimizing the maximum loss within each partitioned set, we can improve the model performance on classes with fewer samples. This helps prevent the model from becoming overly confident and predicting all labels as those of the majority class, and improve the model robustness.

To smooth the objective function $\Phi = \max_{1 \leq k \leq K} \mathcal{L}_k(\theta)$, $\mathcal{L}_k(\theta) = \frac{1}{m_k} \sum_{i=1}^{m_k} \mathcal{L}(f_{\theta}(x_{ki}), y_{ki})$ in the optimization process, $\mathcal{L}_k$ at θ_b is linearized to smooth the maximum function and the convex approximation of Φ can be obtained as follows:

$$\hat{\Phi}(\theta) = \max_{1 \leq k \leq K} \{ \mathcal{L}_k(\theta_b) + \langle \nabla \mathcal{L}_k(\theta_b), \theta - \theta_b \rangle \}. \quad (17)$$

Then we can find the descent direction by minimizing $\hat{\Phi}$. Since this objective function is not strictly convex with respect to θ , the global optimum cannot be achieved. Therefore, to obtain a stable optimization direction, a regularization term $\Delta_{\theta} = \|\theta - \theta_b\|$ is introduced and the problem is equal to:

$$\begin{aligned} \min_{\Delta_{\theta}, \delta} \quad & \|\Delta_{\theta}\| + \delta \\ \text{s.t.} \quad & \mathcal{L}_k(\theta_b) + \langle \nabla \mathcal{L}_k(\theta_b), \Delta_{\theta} \rangle \leq \delta, \quad \forall 1 \leq k \leq K. \end{aligned} \quad (18)$$

In this paper, we choose the L_2 norm and then the computed gradient descent direction at step b is a linear combination of the gradients on all subgroups, which is:

$$\Delta_{\theta_b} = \left\{ -\mathbf{g}_b^T \lambda \mid \sum_{k=1}^K \lambda_k = 1, \lambda_k \geq 0 \right\}, \quad (19)$$

where $\mathbf{g}_b$ is the gradient $\nabla_{\theta} \mathcal{L}$ and λ is the Lagrange multiplier.

By combining the aforementioned algorithm and the gradient descent algorithm to update our model parameters, the process with our structure-based robust fractal graph neural network for molecular property prediction can be summarized in the Algorithm 2.

IV. EXPERIMENTS

In this section, we present seven benchmark datasets, give the implementation details of experiments and discuss the obtained results on different datasets. In addition, we also present the comparison results between our model and the multi-modal models, as well as the results of ablation study.

Algorithm 2: Structure-Based Robust Fractal Graph Neural Network for Molecular Property Prediction

Input: Model f and its parameters θ , training epoch T , molecular dataset $\{(x_1, y_1) \dots (x_n, y_n)\}$, learning rate $\alpha \in \mathbb{R}^+$, number of batch B , loss function $\mathcal{L}$;

Output: The predicted results $\hat{y}$;

```

1: Pre-training MFBERT on unlabeled data;
2: Dataset partition and data processing;
3: while  $t < T$  do
4:   for  $b = 1$  to  $B$  do
5:     Calculate ECFPs from SMILES;
6:     Embedding ECFPs with pre-trained MFBERT;
7:     Forward propagation by FGNN and predict the
       molecular property  $\hat{y}_i = f_{\theta}(\mathbf{x}_i)$ ;
8:     Calculate the loss function
        $\mathcal{L}_k = \frac{1}{m_k} \sum_{i=1}^{m_k} \mathcal{L}(\hat{y}_{ki}, y_{ki})$ ;
9:     Calculate the gradient  $\mathbf{g}_b$ ;
10:    Solve Lagrange multiplier  $\lambda \in \mathbb{R}^K$ :
        $\min_{\lambda} (\frac{1}{2} \lambda^T \mathbf{g}_b \mathbf{g}_b^T \lambda - \mathcal{L}^T \lambda)$ 
       s.t.  $\sum_{i=1}^K \lambda_i = 1, \lambda_i \geq 0$ .
11:     $\Delta_{\theta_b} \leftarrow -\mathbf{g}_b^T \lambda$ ;
12:     $\theta_{b+1}^t = \theta_b^t + \alpha_t \Delta_{\theta_b}$ ;
13:   end for
14:   Learning rate update:  $\alpha_{t+1} = \alpha_t * 0.95$ ;
15:    $t = t + 1$ ;
16: end while
17: Run our model  $f_{\theta}$  on the test set and return predicted
    results  $\hat{y}$ .
```

A. Datasets

Firstly, the unlabeled molecular corpus is necessary for pre-training MFBERT in order to acquire the molecular ECFP embedding. The ZINC dataset [42], [43], [44] provided by the Irwin and Shoichet Laboratories, is used to pre-train our MFBERT model in this paper. ZINC is a free database for virtual screening tasks of compounds and can be used to identify and screen small molecules. Then SMILES format provided by this dataset is used and we selected more than 90 million sequences for training.

Then to evaluate the performance of our model in molecular property prediction tasks, we selected seven datasets from MoleculeNet [45] for the experiments, which are:

- **ESOL** dataset is a widely used benchmark for predicting molecular water solubility, which affects the absorption, distribution, metabolism and excretion properties of molecules and is of significant importance in drug development. The dataset contains 1,128 compounds. Each compound is represented by its chemical structure and associated molecular descriptors. The solubility data is expressed as the logarithm of solubility.
- **FreeSolv** dataset is a benchmark dataset specifically designed for predicting molecular hydration-free energies in water, which plays a crucial role in understanding and predicting the molecular solubility and biological activity. This dataset includes 642 molecules.

TABLE I
DETAILS ABOUT SEVEN BENCHMARK DATASETS

Dataset	Category	#Molecules	#Tasks	Task Type
ESOL	Physical chemistry	1128	1	Regression
FreeSolv	Physical chemistry	642	1	Regression
Lipophilicity	Physical chemistry	4200	1	Regression
BACE	Biophysics	1513	1	Classification
MUV	Biophysics	93087	17	Classification
Tox21	Physiology	7831	12	Classification
SIDER	Physiology	1427	27	Classification

- **Lipophilicity** dataset is an important benchmark for predicting the lipophilicity of compounds, which is represented with octanol/water distribution coefficient (logD at pH 7.4). Lipophilicity plays a crucial role in drug design, as it affects a drug's bioavailability and targeting properties. The dataset size is 4,200.
- **BACE** dataset is used to predict the inhibitory activity of compounds against β -secretase 1 (BACE-1). BACE-1 is a crucial target in Alzheimer's disease treatment, making the BACE dataset significant for drug development. The dataset includes 1,513 compounds along with their corresponding binary labels.
- **MUV (Molecular Understanding of Virulence)** dataset includes 17 prediction tasks, encompassing a variety of bioactivity screenings to assess a model's generalization ability across multiple unrelated bioactivity prediction tasks. Each task is a binary classification problem aimed at predicting whether a compound exhibits activity in a specific biological test. The dataset contains 93,087 compound molecules.
- **Tox21** dataset includes results of 7,831 compounds tested for qualitative toxicity across 12 biological targets, including nuclear receptors and stress response pathways.
- **SIDER** dataset is a database of marketed drugs and adverse drug reactions, designed to provide data on the associations between drugs and their side effects. It aims to help researchers predict potential side effects of medications, thereby enhancing drug safety. The dataset contains over 1,400 drugs, categorized into 27 different system organ classes.

Table I summaries the detailed information about the benchmark datasets, including the dataset category, dataset size, the number and type of tasks.

B. Implementation Details

In this paper, for the molecular SMILES, we used the Morgan algorithm to compute the ECFP sequences, which were then input into MFBERT for embedding. We utilized the RDKit tool to generate the molecular 3D conformations based on distance geometry. Then the experimental torsion angle preferences and basic knowledge such as flat rings are imposed. We also generated 18-dimensional molecular edge features, as shown in Table II. Additionally, we calculated the Coulomb matrix based on the 3D coordinates and Gasteiger charges of molecules.

TABLE II
THE COMPOSITION OF MOLECULAR 18-DIMENSIONAL EDGE FEATURES

Edge Feature	Values	Descriptions
Bond Type	[0,1,2,3]	one-hot embedding for the type of the bond
Bond Direction	[0,1,2,3,4,5,6]	one-hot embedding for the direction of the bond
Stereo	[0,1,2,3,4,5]	one-hot embedding for the stereo configuration of the bond
Conjugated	Bool	whether the bond is considered to be conjugated

TABLE III
THE HYPER-PARAMETERS OF OUR PRE-TRAINED MFBERT MODEL

Parameters	Values
Vocabulary size	15676
Learning rate	1e-5
Batch size	64
Max sequence length	200
Dimension of hidden layers	768
Number of hidden layers	6
Number of attention heads	12
Intermediate size	3072
Type of hidden activation	GELU
Radius	1

For a fair comparison with different methods, we select uniform evaluation metric for the results. In this paper, Root Mean Square Error (RMSE) is used as the evaluation metric for regression tasks. Area Under the Receiver Operating Characteristic Curve (ROC-AUC) is chosen as the evaluation criteria of classification tasks.

Our model architecture is divided into two parts: the pre-trained embedding with MFBERT and molecular property prediction with FGNN. Firstly, in order to obtain the molecular representations, the pre-training of MFBERT was performed on ZINC dataset. With the Morgan algorithm, the molecular ECFP sequences are generated from SMILES. In the Morgan algorithm, we use $r = 1$ to computing ECFP, which not only captures information about the current atom but also incorporates the directly adjacent environment, including neighboring atoms and bond types. This effectively encodes local chemical structure information, such as functional groups, which is often critical for molecular property prediction tasks. Additionally, selecting the radius of 1 reduces unnecessary complexity, mitigates over-fitting and minimizes the risk of introducing noise. Table III shows the MFBERT parameters including the vocabulary size, dimension of hidden layers, number of hidden layers and so on. ADAM is chosen as the optimizer to update MFBERT parameters in the pre-training phases. Through pre-training, MFBERT can capture the molecular hidden features and can be used in MPP via fine-tuning.

The hyper-parameters of FGNN model for different datasets, such as the number of network layers and hidden dimensions, are detailed in Table IV. In this phase, we chose ADAM as the optimizer for updating parameters. The initial learning rate is 0.001 with a learning rate decay parameter of 0.95 and weight decay is 0.00001. To enable the FGNN model to effectively capture the spatial structural information of molecules and achieve a good initialization, we pre-trained our FGNN model

using trusted conformations from the QM9 dataset and inherited the weights from the pre-trained FGNN model for molecular property predictions.

Our training process is divided into two stages. Firstly, ADAM optimizer combined with W-DRO algorithm is used to update the parameters of FGNN and the parameters of MFBERT are frozen. Then we fine-tune the parameters of MFBERT and FGNN model with our optimization algorithm. In this paper, our method is implemented in Python 3.8 with PyTorch 1.8.1 and the experiments are conducted on five Tesla T4 GPUs. We divide the datasets into training, testing, and validation sets according to the setup in [45] and report the results from 10 independent runs.

C. Regression Results

To verify the effectiveness of our structure-based robust fractal graph neural network with MFBERT for molecular property prediction, we first conducted the experiments on three regression datasets, including ESOL, FreeSolv and Lipophilicity. We compare our method with several methods, including N-GramRF [46], GIN [15], PretrainGNN [47], D-MPNN [48], GraphMVP [49], GEM [32], 3D Infomax [33], SPM [50] and ChemBERTa-2 (MTR-77M) [28], using RMSE as the evaluation metric. A smaller RMSE indicates better accurate prediction.

As shown in the Table V, compared to the other methods, our method achieved the lowest RMSE values in all three regression prediction tasks across the datasets, with RMSE of 0.563, 1.157 and 0.612, respectively. Specifically, our method reduced RMSE by at least 0.235 on the ESOL dataset, 0.462 on the FreeSolv dataset, and 0.017 on Lipophilicity dataset. These results demonstrate the effectiveness of our MPP method. Furthermore, compared to the N-GramRF and ChemBERTa-2 (MTR-77M) models, our method achieved smaller prediction errors, highlighting the importance of incorporating neighborhood information for atoms through ECFPs during the pre-training phase. Additionally, in comparison with the graph-based GIN and PretrainGNN models, our results obtained the more accurate prediction results by introducing 3D characteristics. Compared with the existing models that also utilize 3D structures, such as GraphMVP and 3D Infomax, our method still yielded better predictive performance, further validating its effectiveness.

Fig. 6 presents one experimental prediction result of our method and the residual distributions for the three datasets. Firstly, the fitting plots indicate that our method accurately predicts molecular properties. The coefficient of determination (R^2) is used to evaluate the prediction performance and the values are 0.932, 0.898, and 0.753 for three datasets, respectively, which demonstrates the precision of our predictions. Then by analyzing the residual distribution, we observe that the mean residuals of our method are centered around zero. Furthermore, the residual histogram reveals that most residuals are concentrated near zero, exhibiting an approximately bell-shaped curve, which aligns with the assumption of a normal distribution. This indicates that the prediction errors of our method do not show significant systematic bias. The symmetry of histograms suggests similar frequencies of positive and negative errors, further supporting the accuracy of our method.

TABLE IV
THE HYPER-PARAMETERS OF OUR FRACTAL NEURAL NETWORK MODEL FOR MOLECULAR PROPERTY PREDICTION ON SEVEN DATASETS

Parameters \ Dataset	ESOL	FreeSolv	Lipophilicity	BACE	Tox21	SIDER	MUV
Max sequence length	300	50	300	150	100	150	80
Batch size	96	48	48	64	96	64	64
Train epoch	200	200	500	300	300	300	300
Number of FGNN blocks	2	1	1	1	2	1	2
Number of FGNN channels	2	2	2	2	2	2	2
Hidden dimension of FGNN	48	32	64	48	64	64	64
Dropout of FGNN	0.5	0.5	0.5	0.5	0.5	0.5	0.5
Dropout path of FGNN	0.15	0.15	0.15	0.15	0.15	0.15	0.15
Hidden dimension of MLP	48	32	64	48	64	64	64
Cutoff value in RBF	5	5	5	5	5	5	5
Number of RBF kernels	100	100	100	100	100	100	100

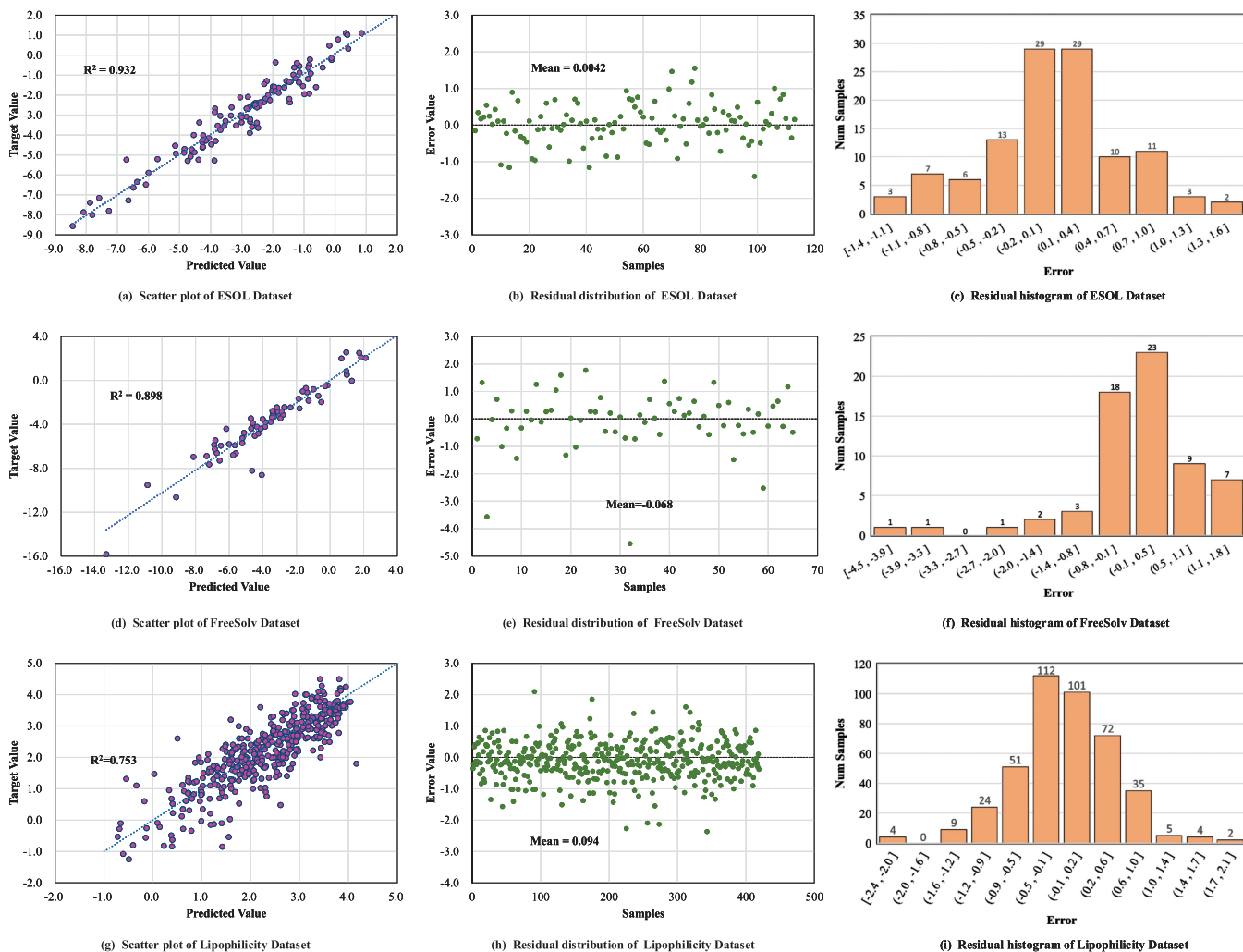


Fig. 6. The scatter plots, residual distributions and residual histograms for ESOL, FreeSolv and Lipophilicity datasets.

D. Classification Results

To further validate the effectiveness of the proposed method, we also conducted experiments on four classification datasets, which are BACE, MUV, Tox21 and SIDER datasets. For the classification tasks, in addition to the methods N-GramRF [46], GIN [15], PretrainGNN [47], D-MPNN [48], GraphMVP [49],

GEM [32], 3D Infomax [33] and ChemBERTa-2 (MTR-77M) [28], we have also included additional baselines for comparison, such as GAL 30B [51], GAL 120B [51], GAL 1.3B [51], GAL 6.7B [51], GAL 125M [51], 3D PGT [52], and MolXPT [53]. ROC-AUC is used as the evaluation metric, which is defined as the area enclosed by the ROC curve and the horizontal axis. The ROC-AUC value ranges from 0.5 to 1.0. The

TABLE V

THE RESULTS OF OUR METHOD COMPARED WITH THE EXISTING METHODS ON THREE REGRESSION DATASETS, INCLUDING ESOL, FREE SOLV AND LIPOPHILICITY DATASETS

Datasets Models	ESOL	FreeSolv	Lipophilicity
N-GramRF	-	2.688 (-)	0.812 (-)
GIN	0.885 (0.051)	<i>1.619 (0.202)</i>	-
PretrainGNN	-	2.764 (-)	0.739 (-)
D-MPNN	1.050 (0.008)	2.082 (0.082)	0.683 (0.016)
GraphMVP	1.029 (0.033)	1.893 (0.063)	0.681 (0.010)
GEM	<i>0.798 (0.029)</i>	1.877 (0.097)	0.660 (0.008)
3D Infomax	0.894 (0.028)	2.337 (0.227)	0.659 (0.012)
SPMM	0.818 (0.008)	1.907 (0.058)	<i>0.629 (0.008)</i>
ChemBERTa-2 (MTR-77M)	0.889 (-)	-	0.798 (-)
Ours	0.563 (0.004)	1.157 (0.033)	0.612 (0.012)

* Bold font indicates the best results, while italic font represents the sub-optimal results.

value of 0.5 indicates that the model’s classification performance is equivalent to random guessing, meaning it lacks classification ability. Conversely, a ROC-AUC value of 1.0 indicates perfect classification accuracy.

We conducted a statistical analysis of the positive and negative samples in four classification datasets. Except for BACE dataset, which exhibits a relatively balanced distribution with 691 positive and 822 negative samples, the other three datasets show a significant imbalance between positive and negative samples, which could impact model performance. In MUV dataset, there are over ten thousand molecular data for each task, but only a few dozen positive samples. Similarly, in the Tox21 dataset, the number of positive samples is significantly lower than the number of negative samples across different sub-tasks. On average, the number of negative samples in the 12 sub-tasks is 20 times greater than the number of positive samples. The SIDER dataset also suffers from imbalance issues, with some sub-tasks having fewer positive samples than negative ones and others the reverse. These situations present significant challenges for molecular property prediction tasks.

Table VI presents the classification results of our method compared to the existing methods. As shown in Table VI, our method achieved the highest ROC-AUC values and the most accurate prediction results on all datasets, highlighting the superior overall performance of our model. On the MUV dataset, compared to existing methods, our model improved the mean ROC-AUC by at least 13.6%, reaching 95.3%. This improvement underscores the effectiveness of our method. Furthermore, our method achieved improvements of 4.1% on BACE dataset, 4.6% on Tox21 dataset and 0.8% on SIDER dataset. Multiple experimental runs also show that the variance of our results remains relatively low, indicating the robustness and stability of our method.

BACE is a single-task dataset and split by scaffold, which can be used to evaluate the model’s robustness and practical applicability. Our method demonstrated commendable performance on this dataset, achieving an accuracy of 0.809, precision of 0.767, recall of 0.896, F1 score of 0.826. These metrics highlight the robustness and reliability of our method in predicting molecular properties. The high precision and recall indicate our model’s strong ability to correctly identify positive instances, while

TABLE VI

ROC-AUC VALUES OF OUR METHOD COMPARED WITH THE EXISTING METHODS ON BACE, MUV, TOX21 AND SIDER DATASETS

Datasets Models	BACE	MUV	Tox21	SIDER
N-GramRF	77.90 (-)	-	74.30 (-)	66.8 (-)
GIN	79.5 (3.4)	-	80.2 (0.2)	59.1 (1.6)
PretrainGNN	84.5 (0.7)	81.3 (2.1)	78.1 (0.6)	62.7 (0.8)
D-MPNN	80.9 (0.6)	-	-	57 (0.7)
GraphMVP	81.2 (0.2)	77.7 (0.6)	74.4 (0.2)	63.9 (1.2)
GEM	85.6 (1.1)	<i>81.7 (0.5)</i>	78.1 (0.1)	67.2 (0.4)
3D Infomax	79.4 (1.9)	-	73.73 (0.69)	58.43 (1.28)
SPMM	84.8 (0.3)	-	-	65.5 (0.9)
ChemBERTa-2 (MTR-77M)	79.9 (-)	-	-	-
GAL 30B	72.7 (-)	-	68.5 (-)	61.3 (-)
GAL 120B	61.7 (-)	-	68.9 (-)	63.2 (-)
GAL 1.3B	57.6 (-)	-	60.6 (-)	54.0 (-)
GAL 6.7B	58.4 (-)	-	63.9 (-)	55.9 (-)
GAL 125M	56.1 (-)	-	54.3 (-)	55.9 (-)
3D PGT	80.9 (-)	69.4 (-)	78.3 (-)	60.6 (-)
MolXPT	<i>88.4 (1.0)</i>	-	77.1 (0.2)	<i>71.7 (0.2)</i>
Ours	92.5 (0.4)	95.3 (0.4)	84.8 (0.3)	72.5 (0.3)

* Bold font indicates the best results, while italic font represents the sub-optimal results.

* The values are presented in percentage format (%).

the accuracy and F1 score reflect its balanced performance in both identifying and classifying molecular properties accurately. Overall, the results demonstrate the effectiveness of our method and its potential for practical applications in molecular property prediction.

Tox21, MUV, and SIDER are multi-label datasets. Fig. 7 illustrates the performance of our method for each label task and different label numbers correspond to different sub-tasks within these datasets. We not only achieved an improvement in the overall average results, but also maintained consistently strong performance across each classification task. Our method does not exhibit poor results in any individual category, highlighting its robustness and effectiveness in multi-class classification scenarios. This demonstrates the model’s ability to generalize well and deliver reliable predictions on the diverse tasks.

E. Comparison Results With Multi-Modal Models

Since our structure-based robust fractal graph neural network with MFBERT in this paper uses molecular 1D ECFP representation and incorporates molecular graphs and 3D conformations in predictions, we further validate its effectiveness by comparing with the multi-modal models on three representative datasets: the regression task dataset ESOL, the single-task classification dataset BACE and the multi-task classification dataset Tox21. The selected multi-modal model baselines include SGGRL [35] and PremuNet [34]. There are two different splitting strategies for molecular property prediction tasks. One is random split, which randomly partitions the dataset into training, validation and test sets. The other is scaffold split, which assigns molecules with different scaffolds to training, validation and test sets. The multi-modal models SGGRL and PremuNet adopted different dataset splitting strategies for molecular property prediction tasks. Specifically, SGGRL used the random split for the ESOL and Tox21 datasets and the scaffold split for the BACE dataset, while PremuNet applies the scaffold split for all three datasets.

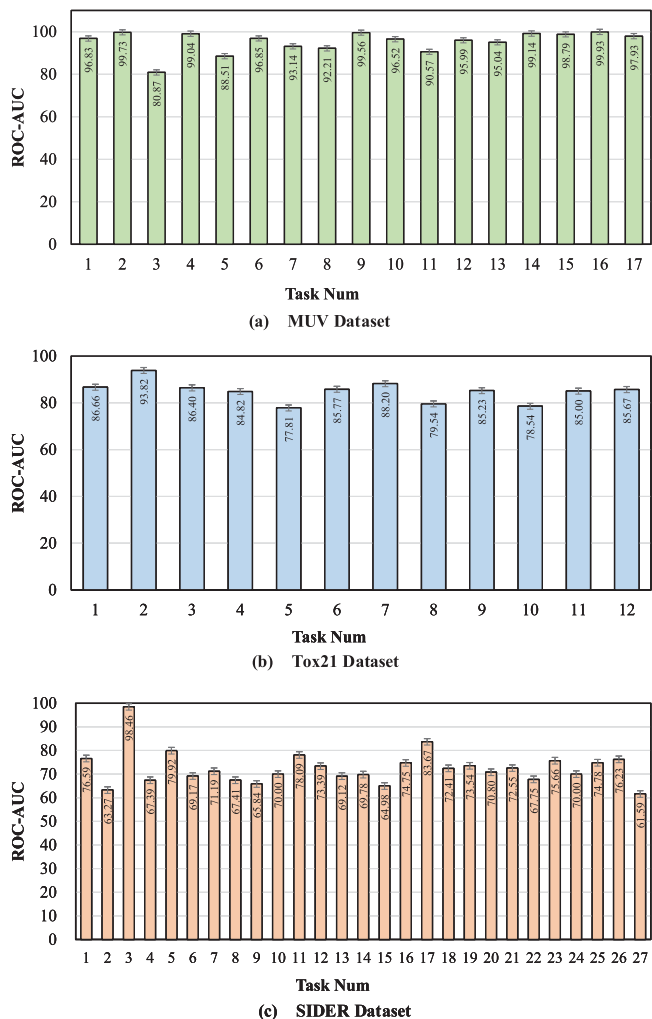


Fig. 7. Histograms of classification results for each task on the MUV, Tox21, and SIDER datasets. ROC-AUC is used as the evaluation metric for our method and the ROC-AUC values are presented as percentages (%).

To ensure a fair and consistent comparison, we conducted experiments following the splitting strategies of each model and compared the results with SGGRL and PremuNet, respectively. The results are shown in Fig. 8.

Firstly, we compared our model with PremuNet, a multi-branch model that integrates 1D SMILES strings, 2D graph representations and 3D conformations. PremuNet consists of two branches: PremuNet-L, which extracts features from SMILES and graph representations, and PremuNet-H, which is a fusion net branch focusing on topological and geometric information. As shown in Fig. 8(a), our method consistently outperforms PremuNet on three datasets. Specifically, on the ESOL dataset, our model achieved the RMSE reduction of 0.025 compared to the PremuNet model. For the classification tasks on the BACE and Tox21 datasets, our model improved the ROC-AUC scores by at least 8.2% and 3.8%, respectively. Additionally, we compared our model with SGGRL, another multi-modal framework that integrates sequence, graph and geometric features for molecular property prediction. As illustrated in Fig. 8(b), our method achieved superior performance on three representative

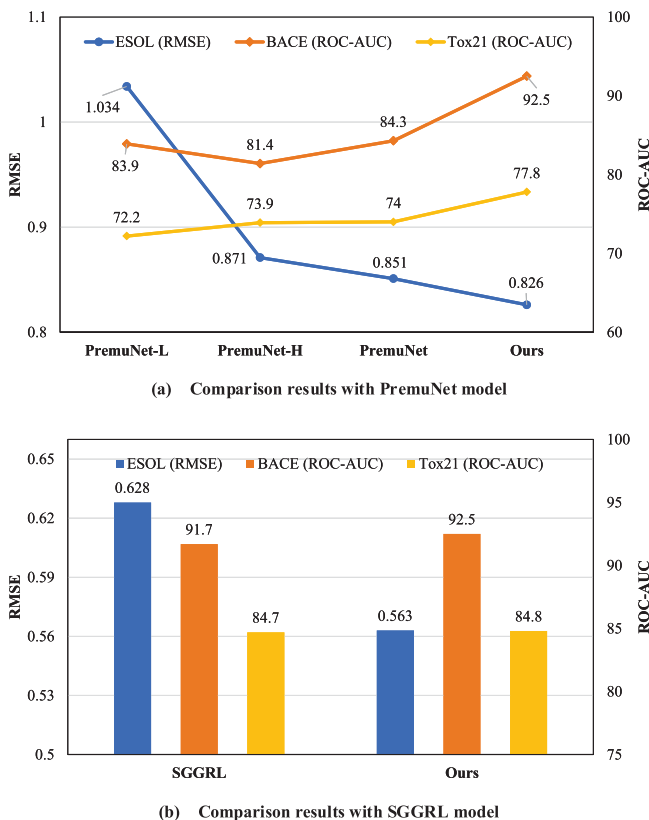


Fig. 8. The comparison results on ESOL, BACE and Tox21 datasets with multi-modal models SGGRL and PremuNet. The left y-axis in the figure represents the evaluation metric RMSE for the regression task (ESOL), where smaller values indicate better performance. The right y-axis represents the evaluation metric ROC-AUC for the classification tasks (BACE and Tox21), where larger values indicate better performance. The ROC-AUC values are presented as percentages (%).

datasets. On the ESOL dataset, our model reduces RMSE from 0.628 to 0.563, while on the two classification datasets (BACE and Tox21), it achieved an average ROC-AUC improvement of 0.45%. Our method consistently outperforms multi-modal models on these tasks, highlighting its superior ability in multi-modal feature learning. These results further validate the effectiveness of our model in predicting molecular properties.

F. Ablation Study

To evaluate the contribution of individual components in our model, we conducted the ablation study. This study aims to understand how each component affects the overall performance of our method. The ablation experiments are conducted on the ESOL regression task and BACE classification task.

We conducted the ablation study by systematically removing or modifying key components of the model. Specifically, we investigated the impact of the pre-trained MFBERT model, pre-trained FGNN on QM9 dataset (FGNN*), FGNN, 3D feature extraction module in FGNN and the improved optimization algorithm. For this study, we created seven variants of our model for ablation experiments: the model with pre-trained FGNN on QM9 dataset (Ours (MFBERT + FGNN* + DRO)), the model without W-DRO algorithm (#(MFBERT + FGNN)), the

TABLE VII
ABLATION STUDY RESULTS ON ESOL AND BACE DATASETS

Models	Datasets	ESOL (RMSE)	BACE (ROC-AUC)
Ours (MFBERT + FGNN*+ DRO)		0.563	92.5
Ours (MFBERT + FGNN + DRO)		0.592	91.9
\$(MFBERT + FGNN)\$		0.604	89.9
\$(Frozen MFBERT + FGNN)\$		0.663	87.3
\$(MFBERT + 2D FGNN)\$		0.676	86.9
\$(MFBERT + FNN)\$		0.720	85.2
\$(Node Feat + FGNN)\$		0.752	82.4
\$(Node Feat + FNN)\$		0.789	75.2

Ours (MFBERT + FGNN+ DRO) represents the model with pre-trained FGNN on QM9 dataset.

*Ours (MFBERT + FGNN + DRO) indicates that FGNN in the model is randomly initialized.

model with frozen MFBERT parameters (#(Frozen MFBERT + FGNN)), FGNN model without the 3D feature extraction module (#(MFBERT + 2D FGNN), the model using MLP to replace graph neural network (#(MFBERT + FNN)), and the model with defined 67-dimensional node features (#(Node Feat + FGNN) and #(Node Feat + FNN)).

Table VII summarizes the results of the ablation study. Firstly, removing the W-DRO algorithm resulted in an increase on RMSE from 0.592 to 0.604 and a decrease in ROC-AUC from 91.9% to 89.9%, indicating its significant contribution to model performance and robustness. Then the effectiveness of using MFBERT pre-trained model is demonstrated by comparing the results of models with defined node features and with the MFBERT embedding, as shown in comparisons of #(Node Feat + FGNN) and #(MFBERT + FGNN), and #(Node Feat + FNN), and #(MFBERT + FNN). Furthermore, frozen the MFBERT parameters led to a reduction of 2.6% in ROC-AUC and an increase of 0.059 in RMSE, which proves that MFBERT model is necessary and fine-tuning can further improve the model's performance on specific tasks.

By comparing the results of using FNN as the predictor with those of using 2D FGNN and our designed FGNN with 3D feature extraction module, for the ESOL task, the error decreased from 0.720, 0.676 to 0.604, and for BACE classification, accuracy improved from 85.2%, 86.9% to 89.9%. This indicates the effectiveness of incorporating the structural features in molecular property prediction tasks, especially the 3D information, which can provide a more comprehensive depict for prediction. The difference between the results of #(Node Feat + FGNN) and #(Node Feat + FNN) also shows the effectiveness of our designed fractal graph neural network.

Finally, we compared the FGNN model with random initialization and pre-trained weights on the QM9 dataset (Ours (MFBERT + FGNN*+ DRO)). As shown in Table VII, on ESOL dataset, RMSE further decreased to 0.563, and on BACE dataset, ROC-AUC further improved to 92.5%, which demonstrates that pre-training the FGNN model can help the model obtain better initialization by learning prior knowledge from the true conformations of molecules and achieve the better performance. These results highlight the positive impact of each component on overall performance. As demonstrated by the ablation study,

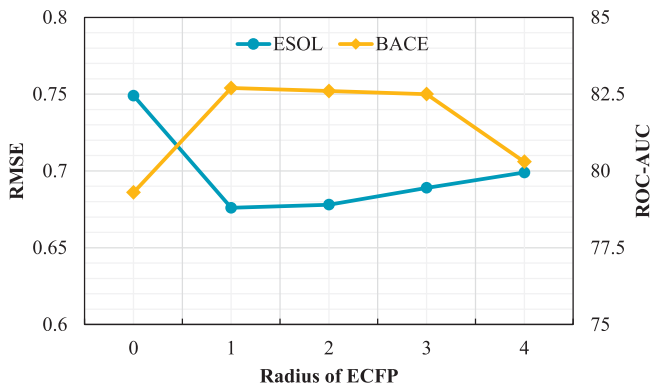


Fig. 9. The experimental results on ESOL and BACE datasets with varying radius r in ECFP. The ROC-AUC values are presented as percentages (%).

the MFBERT model, the pre-trained fractal graph neural network, the 3D feature extraction module in FGNN and W-DRO algorithm are integral to the effectiveness of our model.

We conducted a further analysis of the model's sensitivity to the key hyper-parameter. Since the radius r in ECFP determines the range of initial information capture and significantly impacts model complexity, we performed an in-depth sensitivity analysis on this parameter. We experimented on the ESOL and BACE datasets, setting r to values of 0, 1, 2, 3 and 4. Here, we use ECFP with different r values to process the datasets, applied one-hot encoding and then used the proposed FGNN model for predictions. Fig. 9 shows how the prediction results change with the variation of radius.

As shown in Fig. 9, we can find that the model performs best when the radius parameter is set to 1, which aligns with the chemical intuition. Specifically, with $r = 1$, the model can not only capture the information of the central atom but also include its directly neighboring atoms and bond types. This effectively encodes local chemical structures, such as functional groups, which are crucial in predicting molecular properties. Furthermore, using a smaller radius is sufficient for capturing the structural information of small molecules and helps reduce unnecessary complexity, which can prevent over-fitting and minimize the risk of introducing noise. Conversely, with $r = 0$, the model performs poorly since it can only focus on the current atom and lacks the atomic environmental information. When r is larger, the model complexity increases and may introduce unnecessary noise, which undermines the prediction performance. Therefore, in our structure-based robust fractal graph neural network with MFBERT, we use $r = 1$ for molecular property prediction tasks, which can consider the atom environment information while also reducing the complexity of our model.

V. CONCLUSION

In this paper, we propose a robust fractal graph neural network with a pre-trained MFBERT model for molecular property prediction tasks. We leverage MFBERT to extract molecular structural features from unlabeled molecular data and provide the node representations of molecular graph. Moreover, we

design the fractal graph neural network to incorporate 3D characteristics with a 3D feature extraction module and a coordinate update module, enriching structural details and enhancing predictive performance while ensuring equivariance to molecular rotation, translation, and permutation. By combining W-DRO with the gradient descent algorithm, we further improve the robustness and accuracy of the predictive performance. The results on the ESOL, FreeSolv, and Lipophilicity regression tasks demonstrate that our method achieves the lowest RMSE values compared to existing methods. Additionally, compared with the baseline models, our method achieves the highest classification accuracy on the BACE, MUV, Tox21 and SIDER datasets. Moreover, compared to the multi-modal models, our model also achieves competitive performance. The experimental results demonstrate the effectiveness and robustness of our method in molecular prediction tasks, indicating that it has broad potential applications in bioinformatics and drug discovery. Finally, ablation experiments show that every component of our design is necessary and effective.

REFERENCES

- [1] J. Shen and C. A. Nicolaou, "Molecular property prediction: Recent trends in the era of artificial intelligence," *Drug Discov. Today, Technol.*, vol. 32, pp. 29–36, 2019.
- [2] C. Beaudoin, K. Phalak, and S. Ghosh, "Predicting side effect of drug molecules using recurrent neural networks," *IEEE Trans. Emerg. Topics Comput. Intell.*, early access, Apr., 26, 2024, doi: [10.1109/TETCI.2024.3389778](https://doi.org/10.1109/TETCI.2024.3389778).
- [3] J. C. Slater, "A simplification of the hartree-fock method," *Phys. Rev.*, vol. 81, no. 3, 1951, Art. no. 385.
- [4] D. Van Der et al., "GROMACS: Fast, flexible, and free," *J. Comput. Chem.*, vol. 26, no. 16, pp. 1701–1718, 2005.
- [5] D. A. Case et al., "The Amber biomolecular simulation programs," *J. Comput. Chem.*, vol. 26, no. 16, pp. 1668–1688, 2005.
- [6] L. Tang and Y. Meng, "Data analytics and optimization for smart industry," *Front. Eng. Manage.*, vol. 8, no. 2, pp. 157–171, 2021.
- [7] C. Liu, L. Tang, and J. Liu, "A stacked autoencoder with sparse Bayesian regression for end-point prediction problems in steelmaking process," *IEEE Trans. Automat. Sci. Eng.*, vol. 17, no. 2, pp. 550–561, Apr. 2020.
- [8] Y. Zhang, P. Tiño, A. Leonardi, and K. Tang, "A survey on neural network interpretability," *IEEE Trans. Emerg. Topics Comput. Intell.*, vol. 5, no. 5, pp. 726–742, Oct. 2021.
- [9] E. J. Bjerrum, "Smiles enumeration as data augmentation for neural network modeling of molecules," 2017, *arXiv:1703.07076*.
- [10] M. Hirohara, Y. Saito, Y. Koda, K. Sato, and Y. Sakakibara, "Convolutional neural network based on smiles representation of compounds for detecting chemical motif," *BMC Bioinf.*, vol. 19, pp. 83–94, 2018.
- [11] B. Yu, H. Xie, M. Cai, and W. Ding, "MG-GCN: Multi-granularity graph convolutional neural network for multi-label classification in multi-label information system," *IEEE Trans. Emerg. Topics Comput. Intell.*, vol. 8, no. 1, pp. 288–299, Feb. 2024.
- [12] H. Zhu et al., "Graph structure enhanced pre-training language model for knowledge graph completion," *IEEE Trans. Emerg. Topics Comput. Intell.*, vol. 8, no. 4, pp. 2697–2708, Aug. 2024.
- [13] T. N. Kipf and M. Welling, "Semi-supervised classification with graph convolutional networks," in *Proc. Int. Conf. Learn. Representations*, 2017. [Online]. Available: <https://openreview.net/forum?id=SJU4ayYgl>
- [14] P. Veličković, G. Cucurull, A. Casanova, A. Romero, P. Lio, and Y. Bengio, "Graph attention networks," in *Proc. Int. Conf. Learn. Representations*, 2018. [Online]. Available: <https://openreview.net/forum?id=rJXMpikCZ>
- [15] K. Xu, W. Hu, J. Leskovec, and S. Jegelka, "How powerful are graph neural networks?," in *Proc. Int. Conf. Learn. Representations*, 2019. [Online]. Available: <https://openreview.net/forum?id=ryGs6iA5Km>
- [16] K. Bose and S. Das, "Can graph neural networks go deeper without over-smoothing? yes, with a randomized path exploration!" *IEEE Trans. Emerg. Topics Comput. Intell.*, vol. 7, no. 5, pp. 1595–1604, Oct. 2023.
- [17] D. Rogers and M. Hahn, "Extended-connectivity fingerprints," *J. Chem. Inf. Model.*, vol. 50, no. 5, pp. 742–754, 2010.
- [18] F. H. Kasten, "Paul ehrlich: Pathfinder in cell biology 1. chronicle of his life and accomplishments in immunology, cancer research, and chemotherapy 1: Paul ehrlich's recipe for success: "patience, ability, money and luck," *Biotechnic Histochemistry*, vol. 71, no. 1, pp. 2–37, 1996.
- [19] A. C. Brown and T. R. Fraser, "On the connection between chemical constitution and physiological action; with special reference to the physiological action of the salts of the ammonium bases derived from strychnia, brucia, thebaia, codeia, morphia, and nicotia," *J. Anatomy Physiol.*, vol. 2, no. 2, 1868, Art. no. 224.
- [20] C. Hansch, "Quantitative structure-activity relationships and the unnamed science," *Accounts Chem. Res.*, vol. 26, no. 4, pp. 147–153, 1993.
- [21] P. Hohenberg and W. Kohn, "Inhomogeneous electron gas," *Phys. Rev.*, vol. 136, no. 3B, 1964, Art. no. B864.
- [22] V. Svetnik, A. Liaw, C. Tong, J. C. Culberson, R. P. Sheridan, and B. P. Feuston, "Random forest: A classification and regression tool for compound classification and QSAR modeling," *J. Chem. Inf. Comput. Sci.*, vol. 43, no. 6, pp. 1947–1958, 2003.
- [23] R. Rodríguez-Pérez and J. Bajorath, "Evolution of support vector machine and regression modeling in chemoinformatics and drug discovery," *J. Comput.-Aided Mol. Des.*, vol. 36, no. 5, pp. 355–362, 2022.
- [24] R. P. Sheridan, W. M. Wang, A. Liaw, J. Ma, and E. M. Gifford, "Extreme gradient boosting as a method for quantitative structure–activity relationships," *J. Chem. Inf. Model.*, vol. 56, no. 12, pp. 2353–2360, 2016.
- [25] D. Lemm, G. F. Von Rudorff, and O. A. Von Lilienfeld, "Machine learning based energy-free structure predictions of molecules, transition states, and solids," *Nature Commun.*, vol. 12, no. 1, 2021, Art. no. 4468.
- [26] S. Jaeger, S. Fulle, and S. Turk, "Mol2vec: Unsupervised machine learning approach with chemical intuition," *J. Chem. Inf. Model.*, vol. 58, no. 1, pp. 27–35, 2018.
- [27] D. Weininger, "Smiles, a chemical language and information system. 1. introduction to methodology and encoding rules," *J. Chem. Inf. Comput. Sci.*, vol. 28, no. 1, pp. 31–36, 1988.
- [28] W. Ahmad, E. Simon, S. Chithrananda, G. Grand, and B. Ramsundar, "ChemBERTa-2: Towards chemical foundation models," 2022, *arXiv:2209.01712*.
- [29] C. Lu, Q. Liu, C. Wang, Z. Huang, P. Lin, and L. He, "Molecular property prediction: A multilevel quantum interactions modeling perspective," in *Proc. AAAI Conf. Artif. Intell.*, 2019, pp. 1052–1060.
- [30] Z. Zhang, Q. Liu, H. Wang, C. Lu, and C.-K. Lee, "Motif-based graph self-supervised learning for molecular property prediction," *Adv. Neural Inf. Process. Syst.*, vol. 34, pp. 15870–15882, 2021.
- [31] J. Gasteiger, J. Groß, and S. Günnemann, "Directional message passing for molecular graphs," in *Proc. Int. Conf. Learn. Representations*, 2020. [Online]. Available: <https://openreview.net/forum?id=BlEwBxStPH>
- [32] X. Fang et al., "Geometry-enhanced molecular representation learning for property prediction," *Nature Mach. Intell.*, vol. 4, no. 2, pp. 127–134, 2022.
- [33] H. Stärk et al., "3D infomax improves GNNs for molecular property prediction," in *Proc. Int. Conf. Mach. Learn.*, 2022, pp. 20479–20502.
- [34] H. Zhang, J. Wu, S. Liu, and S. Han, "A pre-trained multi-representation fusion network for molecular property prediction," *Inf. Fusion*, vol. 103, 2024, Art. no. 102092.
- [35] Z. Wang, T. Jiang, J. Wang, and Q. Xuan, "Multi-modal representation learning for molecular property prediction: Sequence, graph, geometry," 2024, *arXiv:2401.03369*.
- [36] S. Job, X. Tao, Y. Li, L. Li, and J. Yong, "Topic integrated opinion-based drug recommendation with transformers," *IEEE Trans. Emerg. Topics Comput. Intell.*, vol. 7, no. 6, pp. 1676–1686, Dec. 2023.
- [37] E. De Santis, A. Martino, F. Ronci, and A. Rizzi, "From bag-of-words to transformers: A comparative study for text classification in healthcare discussions in social media," *IEEE Trans. Emerg. Topics Comput. Intell.*, vol. 9, no. 1, pp. 1063–1077, Feb. 2025.
- [38] Y. Li, B. Jiang, H. Chen, and X. Yao, "Symbolic sequence classification in the fractal space," *IEEE Trans. Emerg. Topics Comput. Intell.*, vol. 5, no. 2, pp. 168–177, Apr. 2021.
- [39] G. Larsson, M. Maire, and G. Shakhnarovich, "Fractalnet: Ultra-deep neural networks without residuals," in *Proc. Int. Conf. Learn. Representations*, 2017. [Online]. Available: <https://openreview.net/forum?id=S1VaB4cex>
- [40] A. Naik, D. R. Edla, and V. Kuppli, "A combination of fractalnet and CNN for lung nodule classification," in *Proc. 11th Int. Conf. Comput., Commun. Netw. Technol.*, 2020, pp. 1–7.
- [41] X. Shen, X. Wang, Q. Xu, W. Ge, and X. Xue, "Towards scalable and fast distributionally robust optimization for data-driven deep learning," in *Proc. IEEE Int. Conf. Data Mining*, 2022, pp. 448–457.

- [42] T. Sterling and J. J. Irwin, "Zinc 15–ligand discovery for everyone," *J. Chem. Inf. Model.*, vol. 55, no. 11, pp. 2324–2337, 2015.
- [43] J. J. Irwin and B. K. Shoichet, "Zinc- a free database of commercially available compounds for virtual screening," *J. Chem. Inf. Model.*, vol. 45, no. 1, pp. 177–182, 2005.
- [44] J. J. Irwin, T. Sterling, M. M. Mysinger, E. S. Bolstad, and R. G. Coleman, "Zinc: A free tool to discover chemistry for biology," *J. Chem. Inf. Model.*, vol. 52, no. 7, pp. 1757–1768, 2012.
- [45] Z. Wu et al., "Moleculenet: A benchmark for molecular machine learning," *Chem. Sci.*, vol. 9, no. 2, pp. 513–530, 2018.
- [46] S. Liu, M. F. Demirel, and Y. Liang, "N-gram graph: Simple unsupervised representation for graphs, with applications to molecules," in *Proc. Adv. Neural Inf. Process. Syst.*, 2019, pp. 8466–8478.
- [47] W. Hu et al., "Strategies for pre-training graph neural networks," in *Proc. Int. Conf. Learn. Representations*, 2020. [Online]. Available: <https://openreview.net/forum?id=HJIWWJSFDH>
- [48] K. Yang et al., "Analyzing learned molecular representations for property prediction," *J. Chem. Inf. Model.*, vol. 59, no. 8, pp. 3370–3388, 2019.
- [49] S. Liu, H. Wang, W. Liu, J. Lasenby, H. Guo, and J. Tang, "Pre-training molecular graph representation with 3D geometry," in *Proc. Int. Conf. Learn. Representations*, 2022. [Online]. Available: <https://openreview.net/forum?id=xQe1pOKPam>
- [50] J. Chang and J. C. Ye, "Bidirectional generation of structure and properties through a single molecular foundation model," *Nature Commun.*, vol. 15, no. 1, 2024, Art. no. 2323.
- [51] R. Taylor et al., "Galactica: A large language model for science," 2022, *arXiv:2211.09085*.
- [52] X. Wang, H. Zhao, W.-W. Tu, and Q. Yao, "Automated 3D pre-training for molecular property prediction," in *Proc. 29th ACM SIGKDD Conf. Knowl. Discov. Data Mining*, 2023, pp. 2419–2430.
- [53] Z. Liu et al., "MolXPT: Wrapping molecules with text for generative pre-training," in *Proc. 61st Annu. Meeting Assoc. Comput. Linguist.*, Toronto, Canada, Jul. 2023, pp. 1606–1616. [Online]. Available: <https://aclanthology.org/2023.acl-short.138/>



Yaguo Dong is currently a student with National Frontiers Science Center for Industrial Intelligence and Systems Optimization, Northeastern University, Shenyang, China and Key Laboratory of Data Analytics and Optimization for Smart Industry, Northeastern University, Ministry of Education, China. His research interests include natural language processing, AI-based drug design and optimization methods.



Her research papers have appeared in academic journals such as *IEEE Transactions on Cybernetics*, *IEEE Transactions on Neural Networks and Learning Systems*, and *IEEE Transactions on Systems, Man, and Cybernetics: Systems*.



Lixin Tang (Fellow, IEEE) received the B.Eng. degree in industrial automation, the M.Eng. degree in systems engineering and the Ph.D. degree in control science and engineering from Northeastern University, Shenyang, China, in 1988, 1991, 1996, respectively.

He is a member of Chinese Academy of Engineering, the Director of Key Laboratory of Data Analytics and Optimization for Smart Industry, Ministry of Education, China, and the Director and Chair Professor of National Frontiers Science Center for Industrial Intelligence and Systems Optimization, Northeastern University.

His research interests cover industrial intelligence and systems optimization theories and methods, covering data analytics and machine learning, deep learning and evolutionary learning, reinforcement learning and dynamic optimization, convex and sparse optimization, integer and combinatorial optimization, computational intelligence-based optimization. Meanwhile, he applies the above theories and technologies to engineering applications in steel manufacturing industry, equipment/chip manufacturing industry, energy industry, logistics industry and information industry.

He has published many papers in international journals such as *IEEE Transactions on Evolutionary Computation*, *IEEE Transactions on Cybernetics*, *IEEE Transactions on Control Systems Technology*, *IEEE Transactions on Automation Science and Engineering*, *IEEE Transactions on Neural Networks and Learning Systems*, *IEEE Transactions on Power Systems*, *Operations Research*, *INFORMS Journal on Computing*, et al. His paper published on *IJSE Transactions* received the Best Applications Paper Award of 2017.

He currently serves as an Associate Editor of *IEEE Transactions on Cybernetics*, *IJSE Transactions*, *Journal of Scheduling* and *International Journal of Production Research*. Meanwhile, he is on the Editorial Board of *Annals of Operations Research*, and serves as an Area Editor of the *Asia-Pacific Journal of Operational Research*.